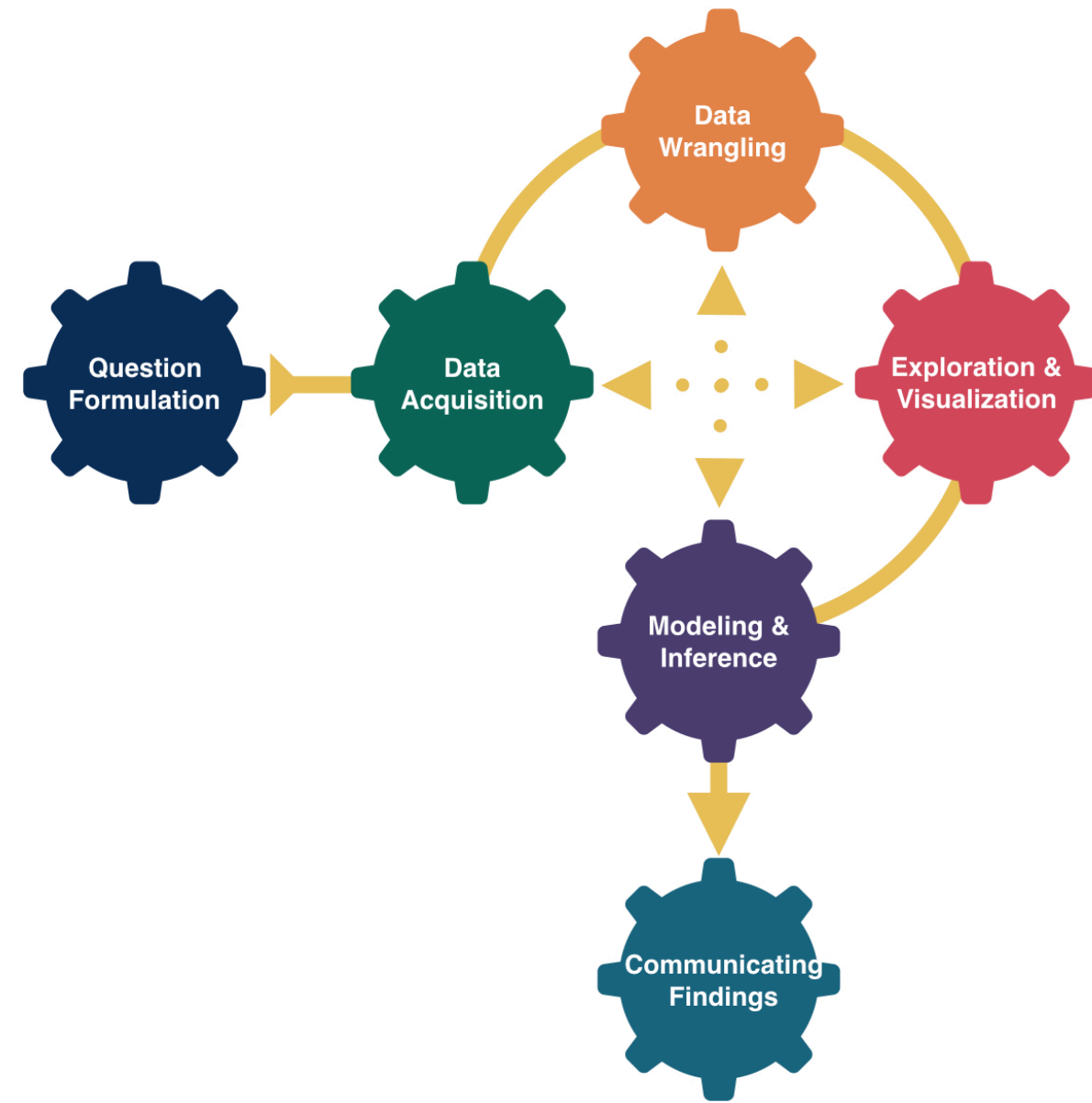


# Inference: Random Variables + P- Value Pitfalls



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DATA 100

Week 10 | Spring 2026

# Announcements

- Exam next week:
  - In-class on Wed (April 8th)
  - Orals throughout the day on Fri (April 10th). **In-person** this time!

## Goals for Today

- Statistical inference zoom out
- Approximating inference using random variables
- A hearty p-values discussion

# Which Flavor of Data Scientist Are You?

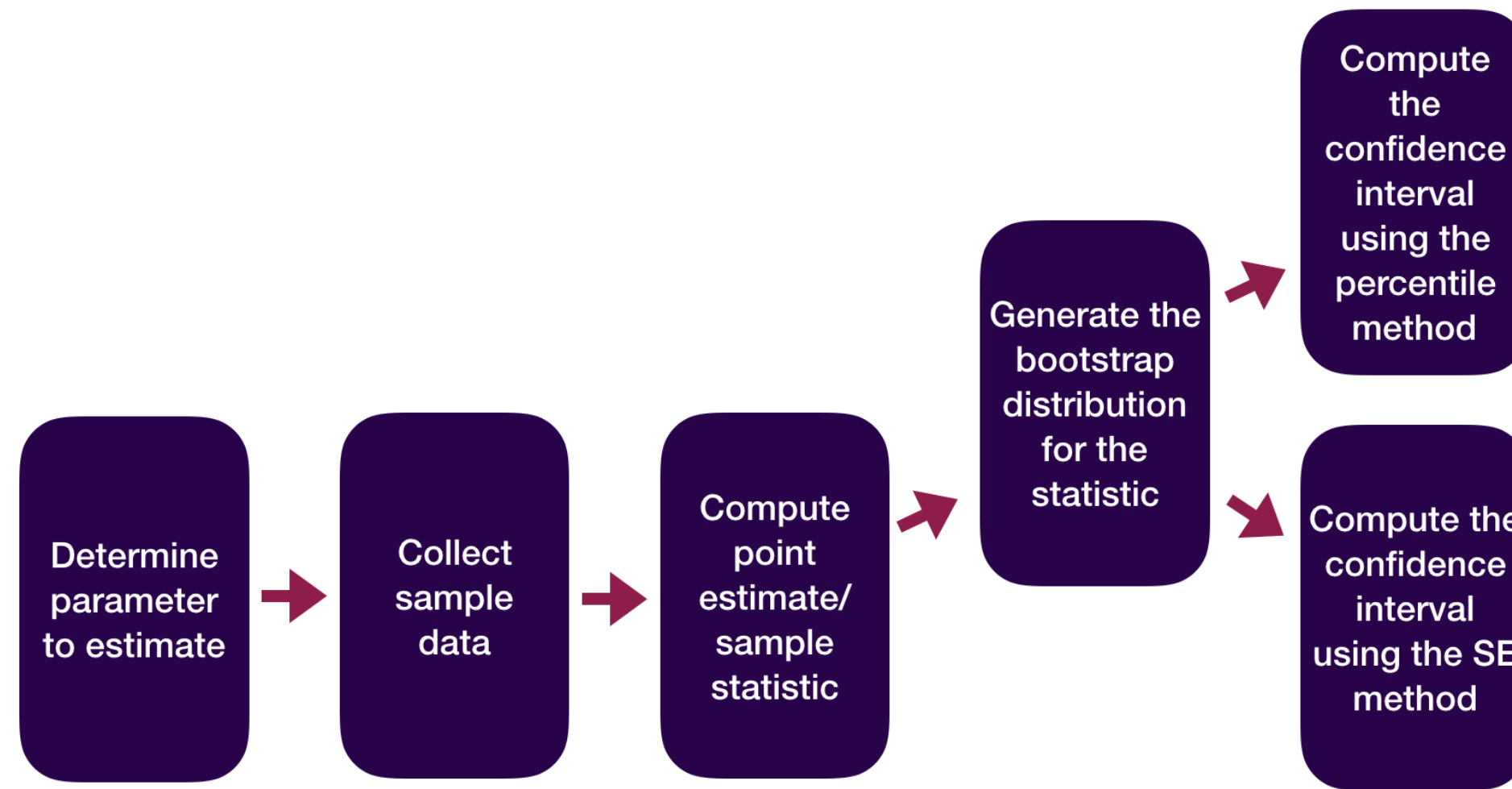
Data Visualizer

Model Builder

Data Wrangler

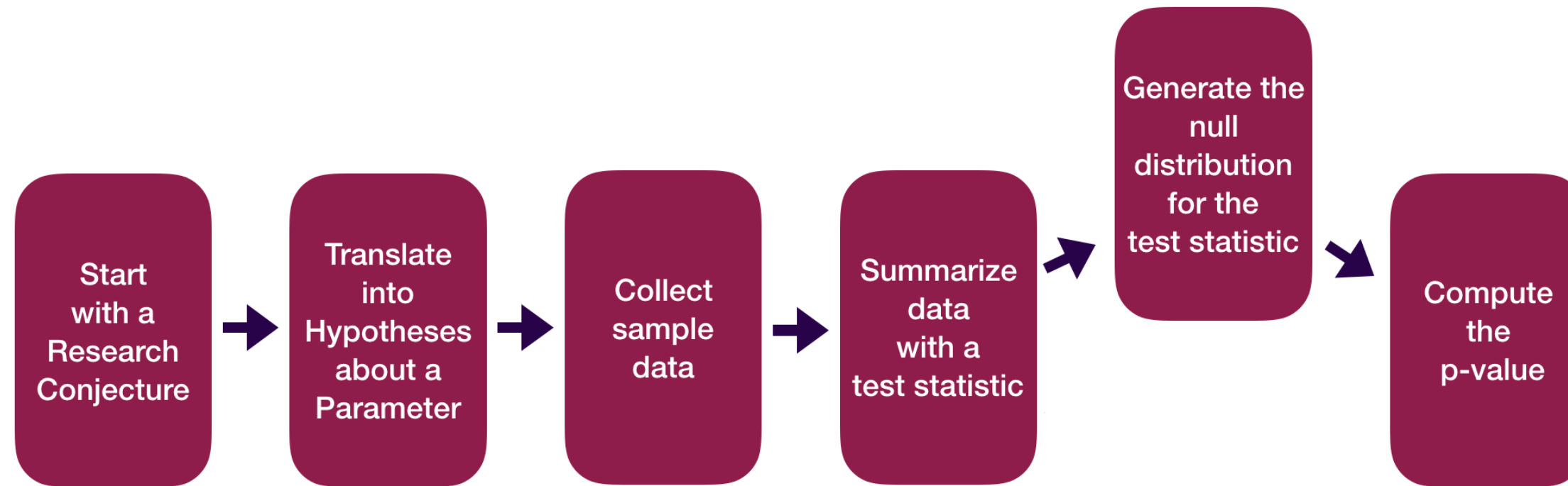
Inferencer

# Statistical Inference Zoom Out – Estimation



**Question:** How did folks do inference before computers?

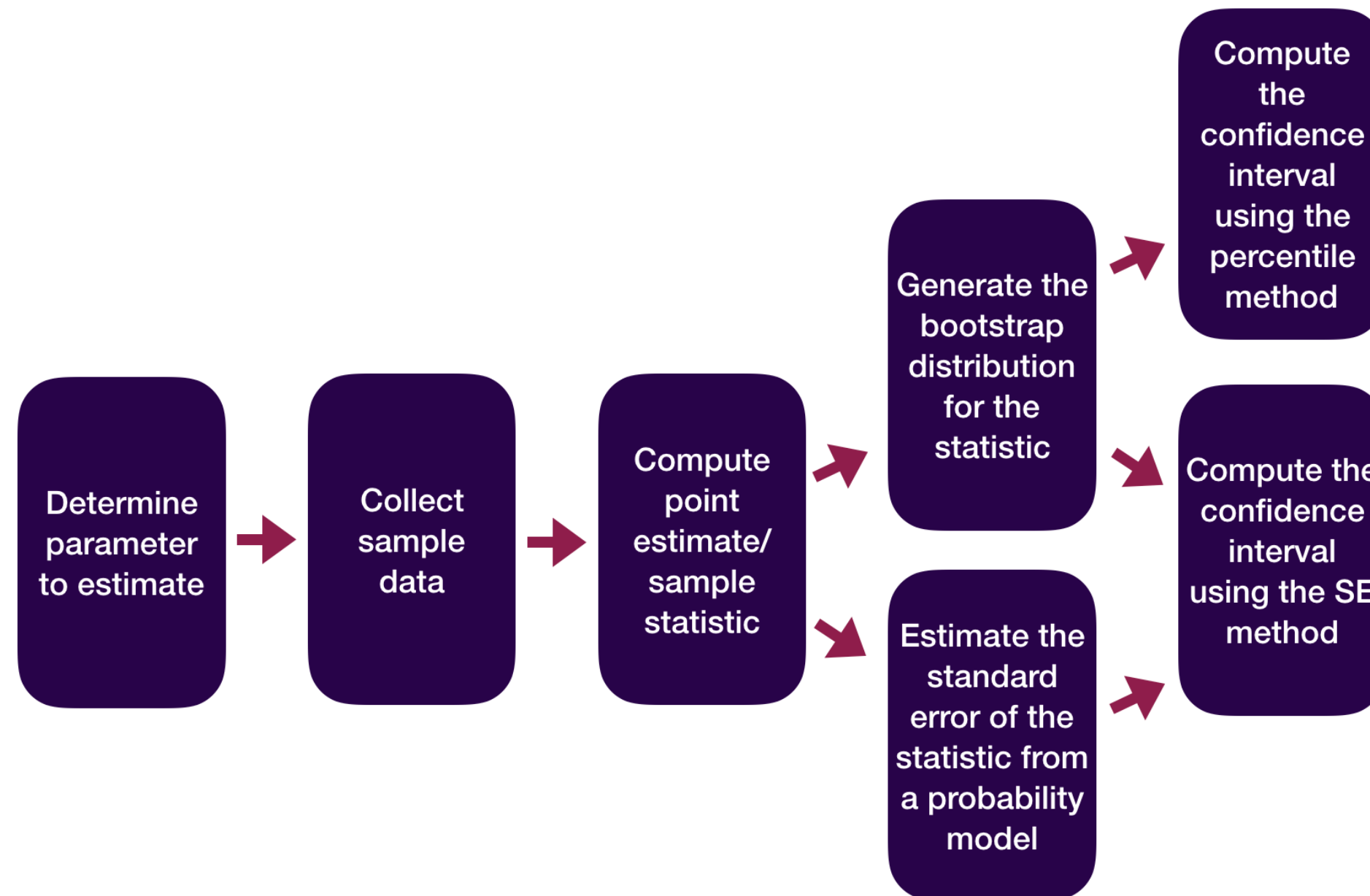
# Statistical Inference Zoom Out – Testing



**Question:** How did folks do inference before computers?

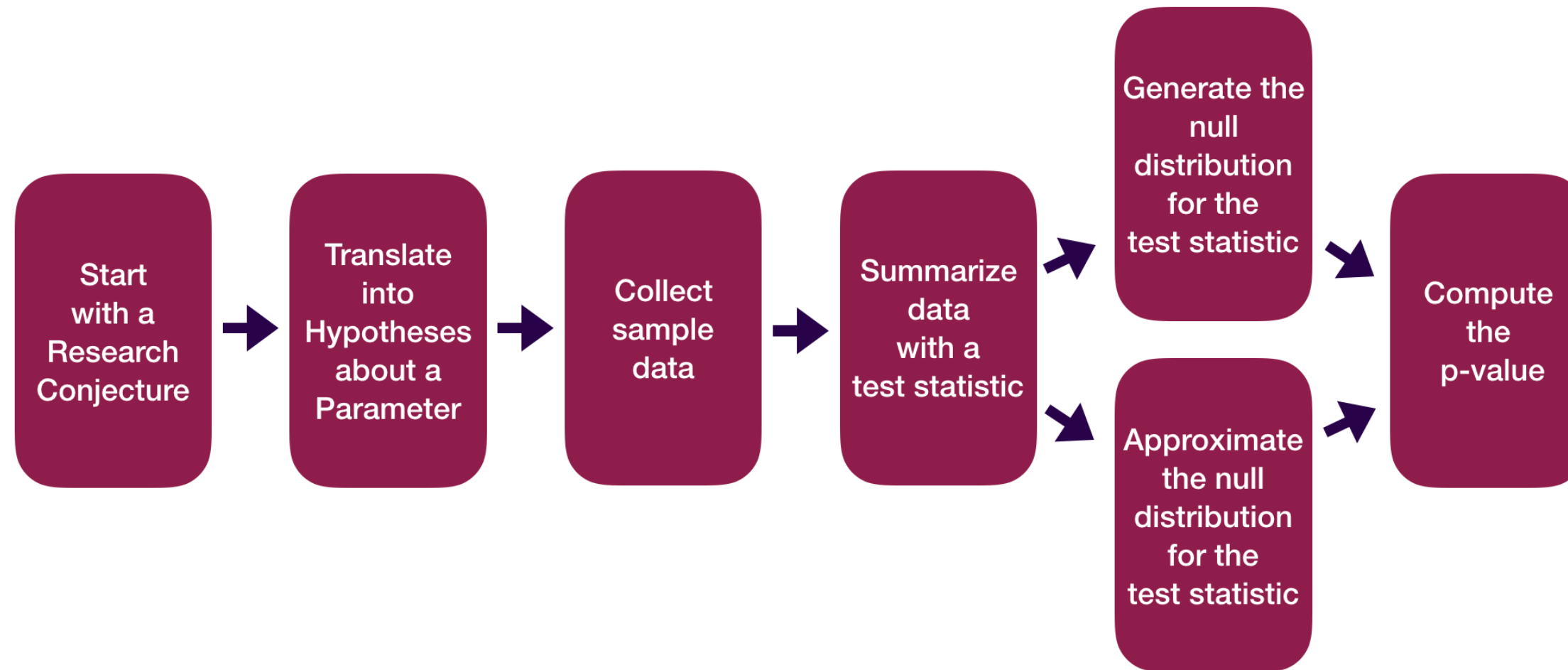


# Statistical Inference Zoom Out – Estimation



**Question:** How did folks do inference before computers?

# Statistical Inference Zoom Out – Testing



**Question:** How did folks do inference before computers?



**It is time to recast some of the sample statistics we have been exploring as random variables!**

# Sample Statistics as Random Variables

**Random variable** (RV) is a random process that **takes on numerical values**.

- Statisticians understand the shape, center, and spread of MANY random variables.

Here are some of the sample statistics we've seen lately:

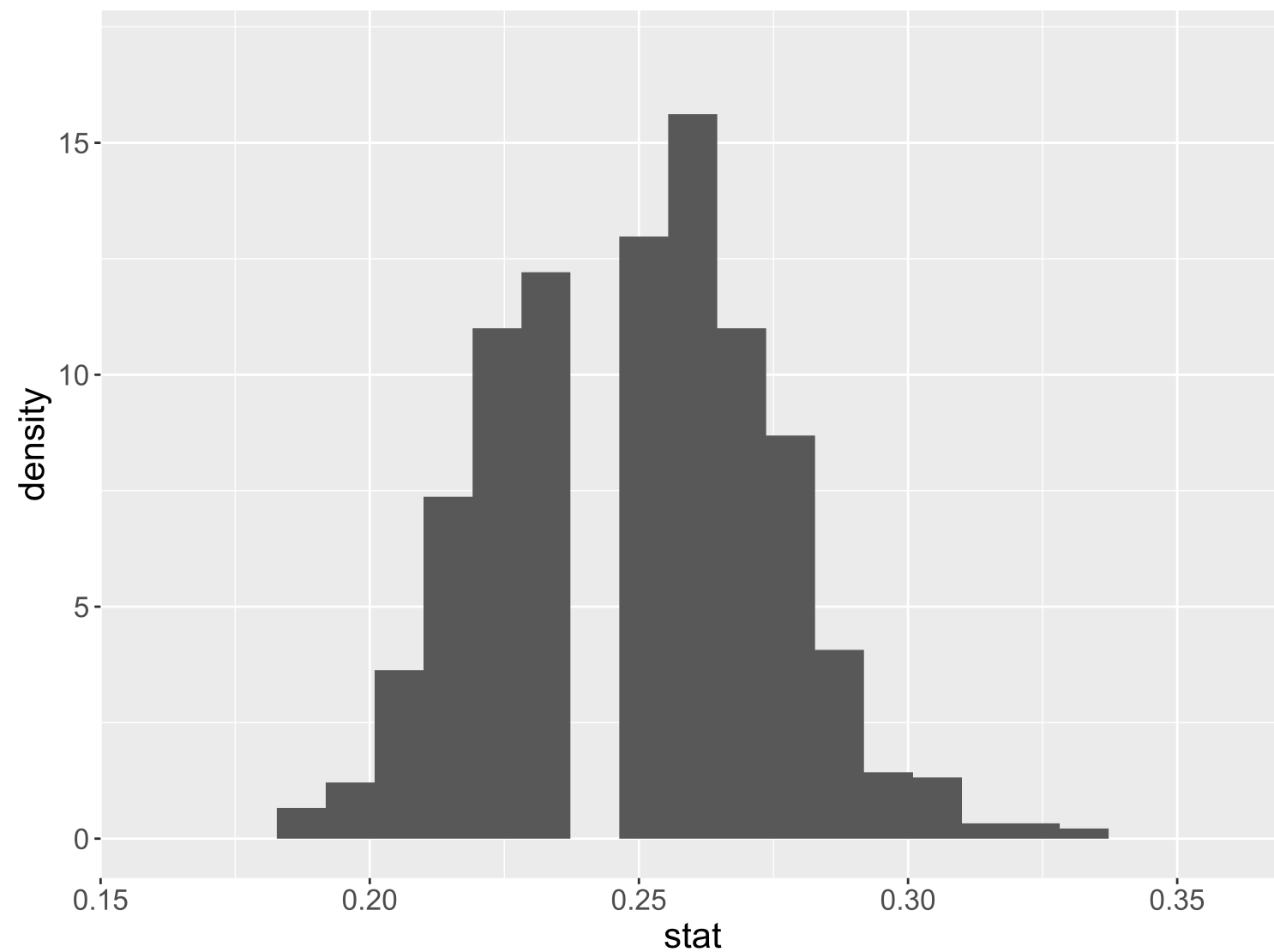
- $\hat{p}$  = sample proportion of correct receiver guesses out of 329 trials
- $\hat{p}_D - \hat{p}_Y$  = difference in sample improvement proportions between those who swam with dolphins and those who did not

Why are these all random variables?

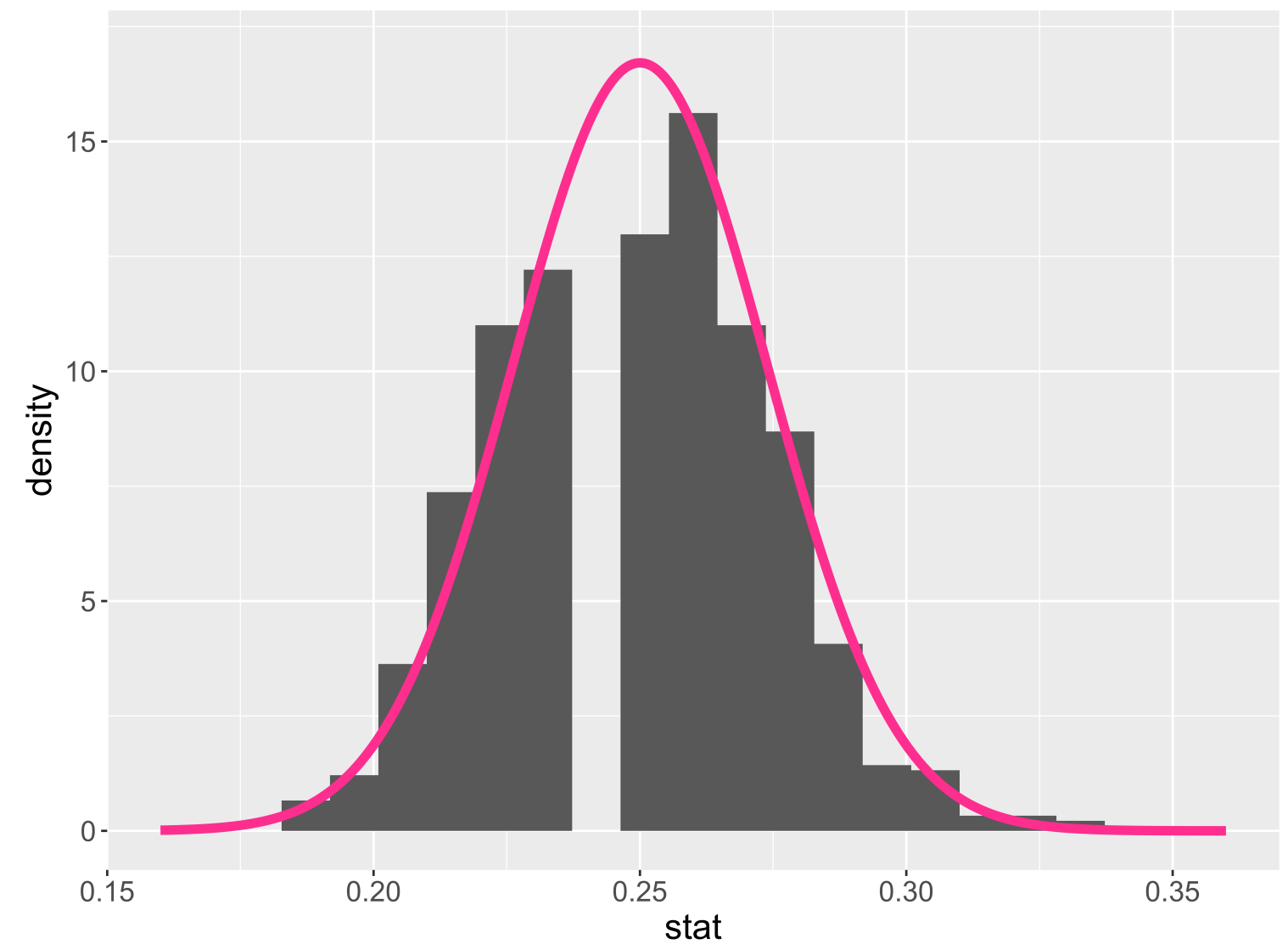
# Approximating These Distributions

- $\hat{p}$  = sample proportion of correct receiver guesses out of 329 trials

We generated its Null Distribution:



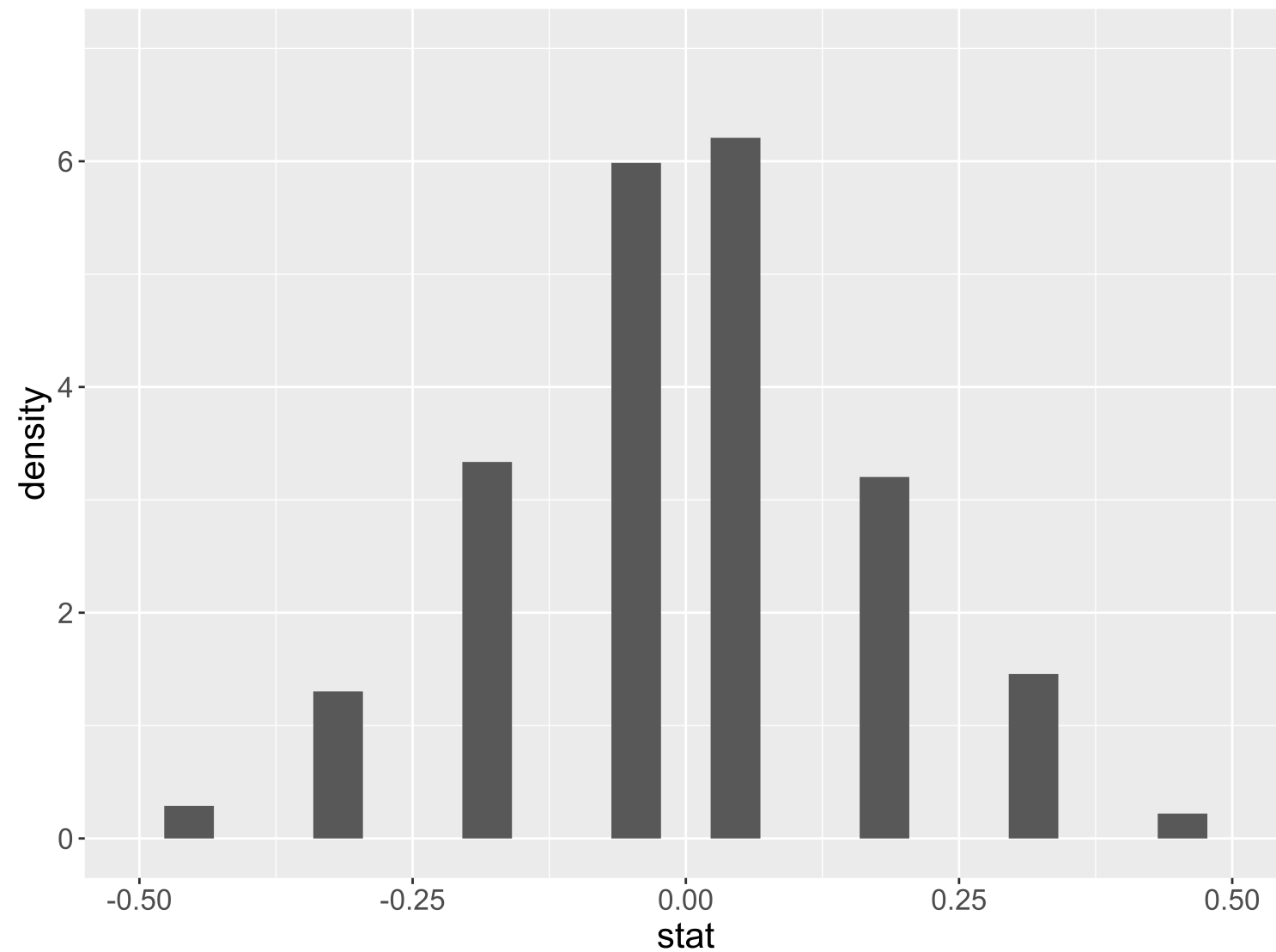
Which is well approximated by the distribution of a Normal random variable.



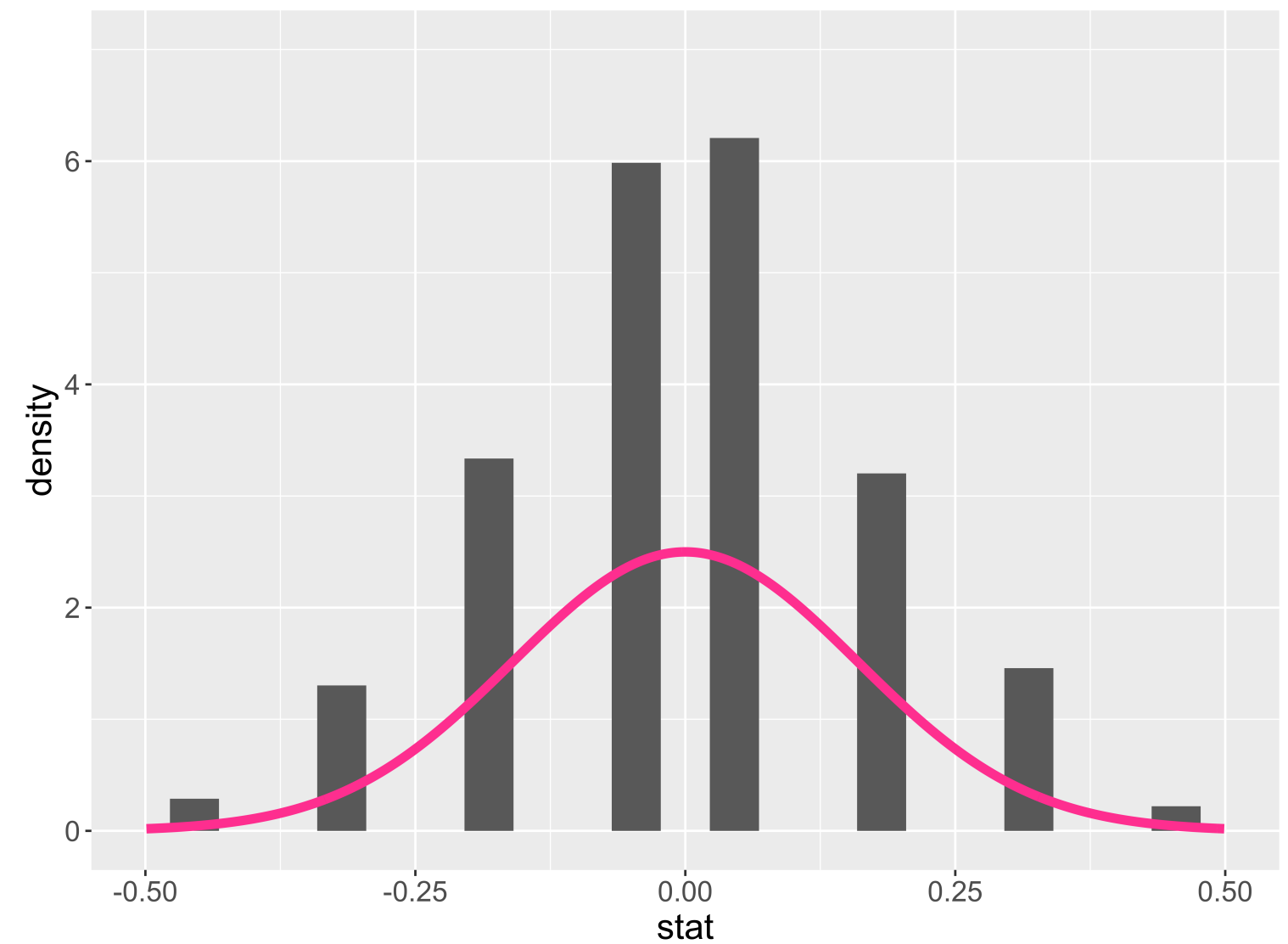
# Approximating These Distributions

- $\hat{p}_D - \hat{p}_Y$  = difference in sample improvement proportions between those who swam with dolphins and those who did not

We generated its Null Distribution:



Which is **kinda somewhat** approximated by the distribution of a Normal random variable.



***“All models are wrong but some are useful.” – George Box***

# Approximating These Distributions

- How do I know **which** random variable is a good approximation for my statistic?
- Once I have figured out the random variable, how do I **use it** to do statistical inference?

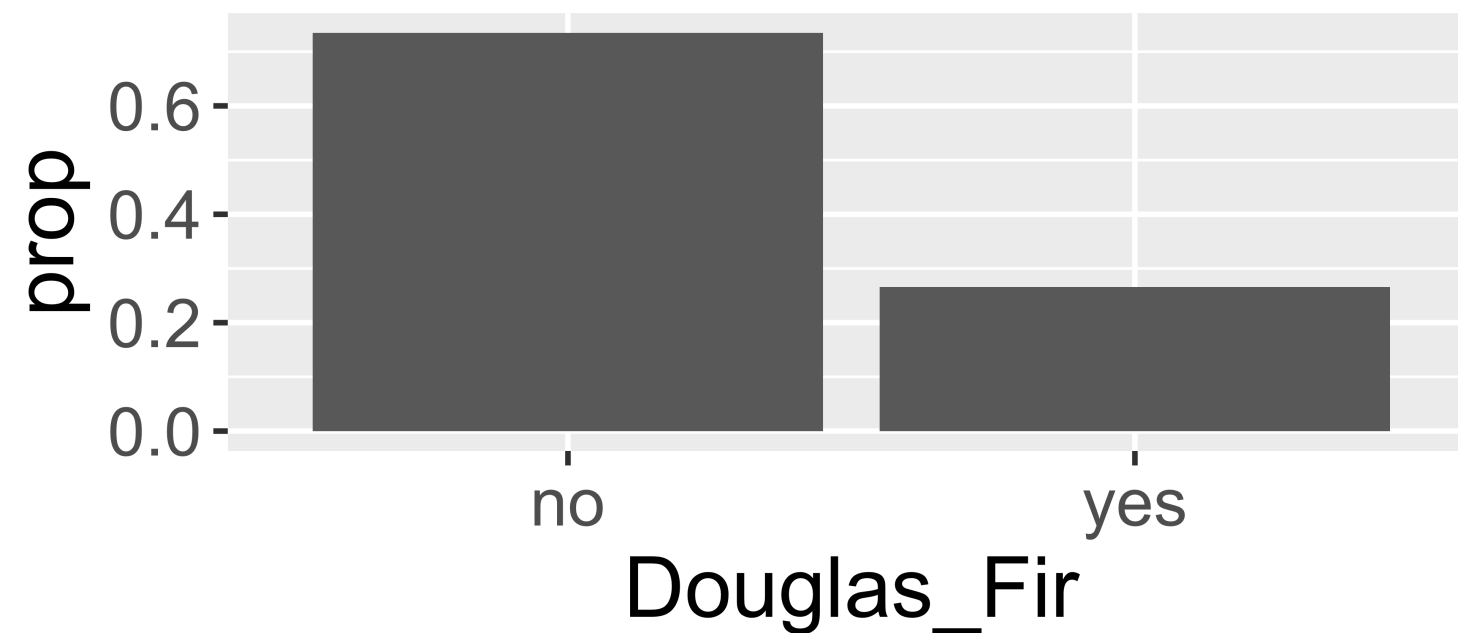


# Central Limit Theorem

**Central Limit Theorem (CLT):** For random samples and a large sample size ( $n$ ), the sampling distribution of many sample statistics is approximately normal.

**Example:** Portland Park Trees

The Population  
Distribution

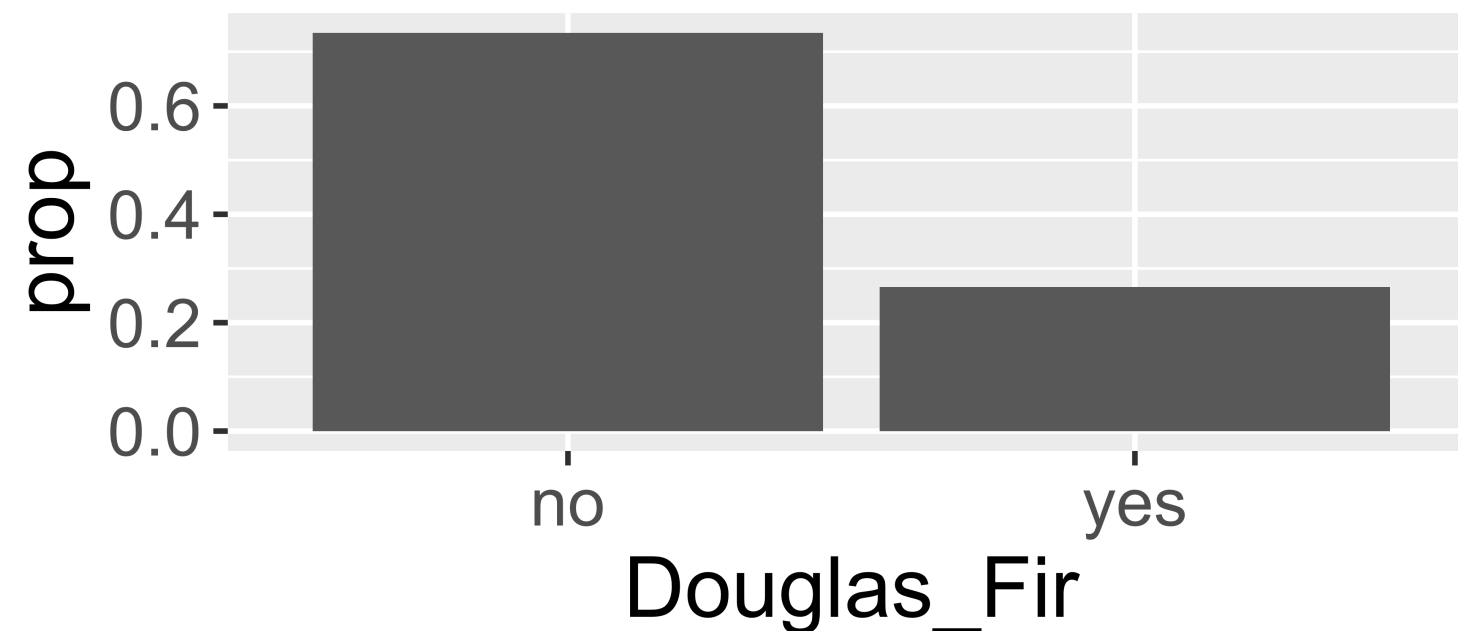


# Approximating Sampling Distributions

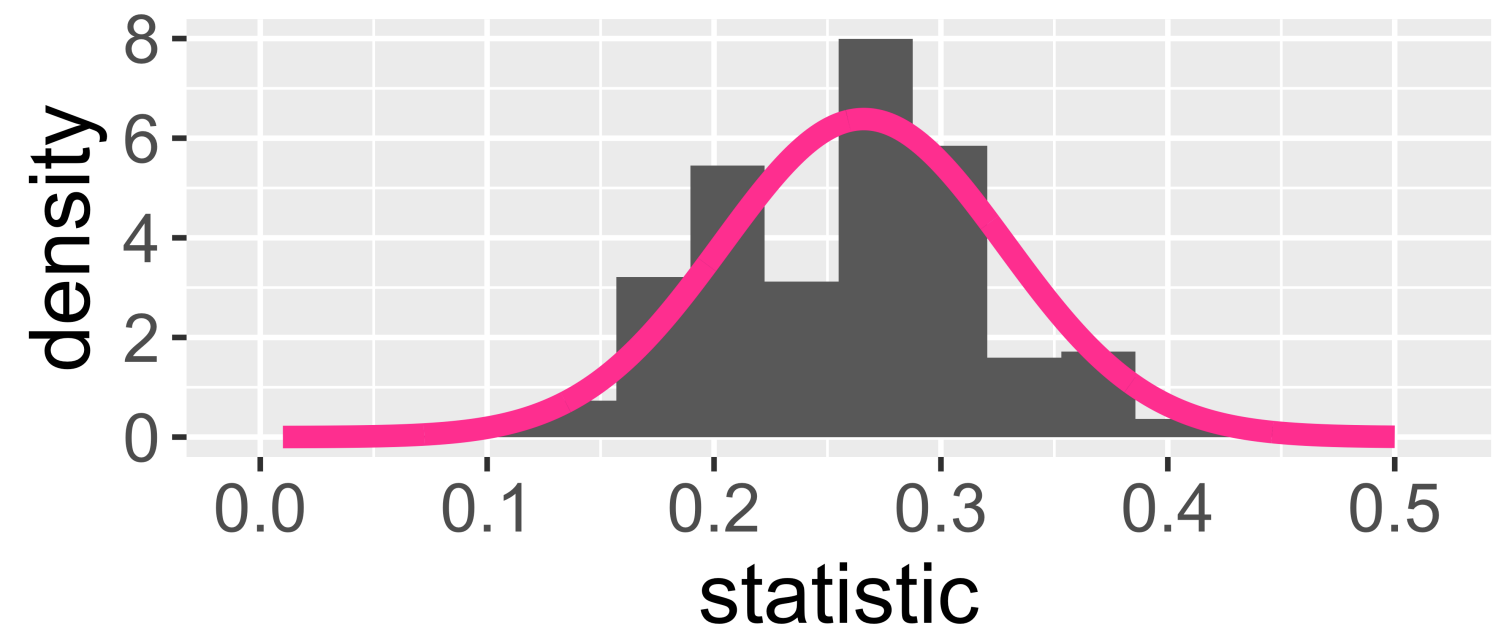
**Central Limit Theorem (CLT):** For random samples and a large sample size ( $n$ ), the sampling distribution of many sample statistics is approximately normal.

**Example:** Harvard Trees

## The Population Distribution



## The Sampling Distribution



- The larger the sample size, the closer the distribution gets to being bell-shaped and symmetric, like a Normal random variable.



Sometimes it is easier to find a  
good random variable  
approximation if we standardize  
our sample statistic first.

# Z-scores

- All of our **test statistics** so far have been **sample statistics**.
- Another commonly used test statistic takes the form of a **z-score**:

$$\text{Z-score} = \frac{X - \text{mean}}{\text{standard deviation}}$$

- Standardized version of the sample statistic.
- Z-score measures how many **standard deviations** the sample statistic is away from its **mean**.

# Z-score Example

- $\hat{p}$  = proportion of Maples in a sample of 50 trees
- Mean of  $\hat{p}$ : 0.138
- Standard deviation of  $\hat{p}$ : 0.049
- Suppose we have a sample where  $\hat{p} = 0.05$ . Then the z-score would be:

$$\text{Z-score} = \frac{0.05 - 0.138}{0.049} = -1.8$$

# Z-score Test Statistics

- A Z-score test statistic is one where we take our original sample statistic and convert it to a Z-score:

$$\text{Z-score} = \frac{\text{statistic} - \text{mean}}{\text{standard deviation}}$$

- Allows us to quickly (but roughly) classify results as unusual or not.
  - $| \text{Z-score} | > 2 \rightarrow$  results are unusual/p-value will be smallish

# Inference based on Random Variable Approximation: One Proportion

Let's consider conducting a hypothesis test for a single proportion:  $p$

**Example:** Bern and Honorton's (1994) extrasensory perception (ESP) studies

```
1 # Construct data frame of sample results
2 esp <- data.frame(guess = c(rep("correct", 106), rep("incorrect", 329 - 106)))
```

```
1 # Theory-based approximation
2 prop_test(esp, response = guess, success = "correct", p = 0.25,
3           z = TRUE, alternative = "greater")
```

```
# A tibble: 1 × 3
  statistic p_value alternative
  <dbl>    <dbl> <chr>
1     3.02 0.00125 greater
```

```
1 # Confidence interval
2 prop_test(esp, response = guess, success = "correct",
3           conf_int = TRUE, conf_level = 0.95)
```

```
# A tibble: 1 × 6
  statistic chisq_df p_value alternative lower_ci upper_ci
  <dbl>    <int>    <dbl> <chr>          <dbl>    <dbl>
1     40.9        1 1.60e-10 two.sided    0.273    0.376
```

When is the random variable approximation reasonable?

# Inference for a Single Mean

**Example:** *Are lakes in Florida more acidic or alkaline?* The pH of a liquid is the measure of its acidity or alkalinity where pure water has a pH of 7, a pH greater than 7 is alkaline and a pH less than 7 is acidic. The following dataset contains observations on a sample of 53 lakes in Florida.

```
1 library(tidyverse)
2 FloridaLakes <- read_csv("https://www.lock5stat.com/datasets1e/FloridaLakes.csv")
```

**Cases:**

**Variable of interest:**

**Parameter of interest:**

**Hypotheses:**



# Inference for a Single Mean: Simulation-Based Approach

```
1 library(infer)
2
3 #Compute obs stat
4 xbar <- FloridaLakes %>%
5   specify(response = pH) %>%
6   hypothesize(null = "point", mu = 7) %>%
7   calculate(stat = "mean")
8 xbar
```

```
Response: pH (numeric)
Null Hypothesis: point
# A tibble: 1 × 1
  stat
<dbl>
1  6.59
```

```
1 # Generate null distribution
2 null_dist <- FloridaLakes %>%
3   specify(response = pH) %>%
4   hypothesize(null = "point", mu = 7) %>%
5   generate(reps = 1000, type = "bootstrap") %>%
6   calculate(stat = "mean")
7
8 pvalue <- null_dist %>%
9   get_p_value(obs_stat = xbar,
10              direction = "both")
11 pvalue
```

```
# A tibble: 1 × 1
  p_value
<dbl>
1  0.024
```

Why are we using `type = "bootstrap"` when constructing a null distribution?!

# Inference for a Single Mean: Random Variable-Based Approach

```
1 t_test(FloridaLakes, response = pH, mu = 7,  
2       alternative = "two-sided")
```

# A tibble: 1 × 7

|   | statistic | t_df  | p_value | alternative | estimate | lower_ci | upper_ci |
|---|-----------|-------|---------|-------------|----------|----------|----------|
|   | <dbl>     | <dbl> | <dbl>   | <chr>       | <dbl>    | <dbl>    | <dbl>    |
| 1 | -2.31     | 52    | 0.0247  | two.sided   | 6.59     | 6.24     | 6.95     |

When is the random variable approximation reasonable?



# Statistical Inference using Random Variables

- We went through random variable-based inference for  $p$  and for  $\mu$ .
- There are similar results for other parameters. But the random variable approximation changes.
  - Will extend beyond inference for 1 variable next time.

# P-Values

# Let's Talk About P-values

- The original intention of the p-value was as an **informal** measure to judge whether or not a researcher should take a second look.
- But to create simple statistical manuals for practitioners, this informal measure quickly became a rule: **“p-value < 0.05” = “statistically significant”**.

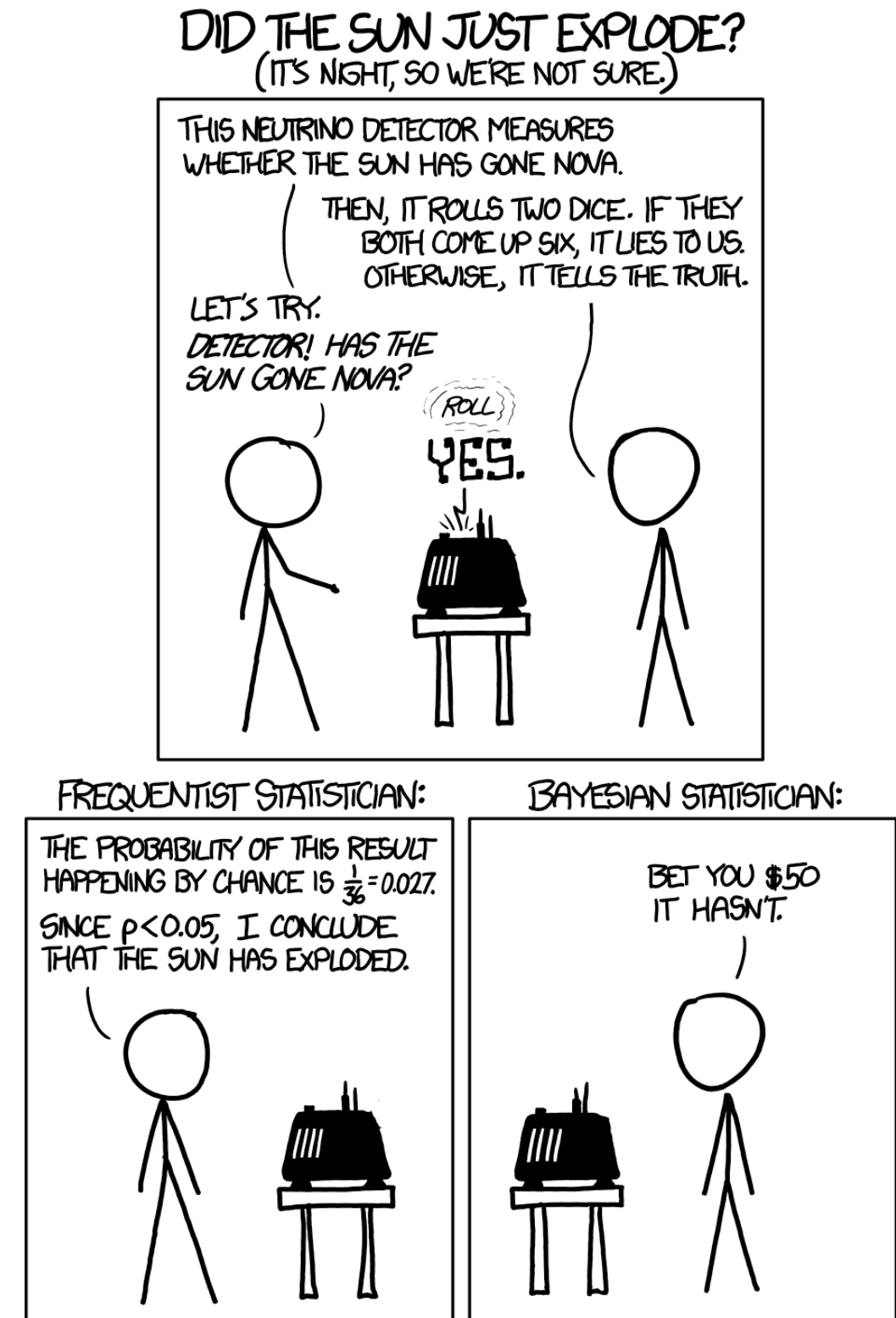
**What were/are the consequences of the “p-value < 0.05” = “statistically significant” rule?**

# Let's Talk About P-values

- **A consequence:** The p-value is often misinterpreted to be the probability the null hypothesis is true.
  - A p-value of 0.003 does not mean there's a 0.3% chance that ESP doesn't exist!
  - By giving people a simple rule, they never learned what the p-value actually measures.

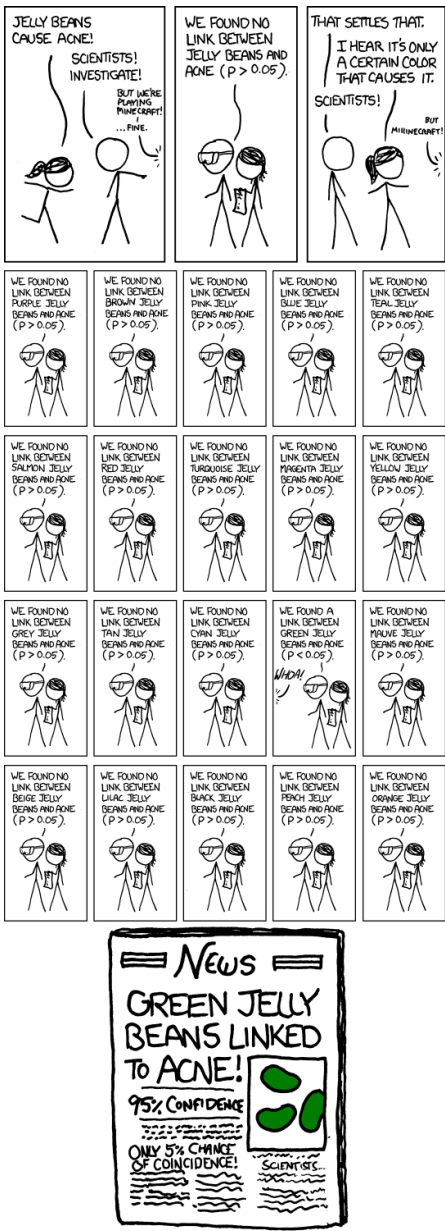
# Let's Talk About P-values

- **A consequence:** Researchers often put too much weight on the p-value and not enough weight on their domain knowledge/the plausibility of their conjecture.
  - Sometimes we give more weight to things we don't understand.
- [xkcd comic](#)



# Let's Talk About P-values

- **A consequence:** P-hacking: Cherry-picking promising findings that are beyond this arbitrary threshold.
- [xkcd comic](#)





# Let's Talk About P-values

- **A consequence:** People conflate *statistical significance* with *practical significance*.

**Example:** A recent *Nature* study of 19,000+ people found that those who meet their spouses online...

- Are less likely to divorce (p-value < 0.002)
- Are more likely to have high marital satisfaction (p-value < 0.001)
- BUT the estimated **effect sizes** were tiny.
  - Divorce rate of 5.96% for those who met online versus 7.67% for those who met in-person.
  - On a 7 point scale, happiness value of 5.64 for those who met online versus 5.48 for those who met in-person.

**Question:** Do these results provide compelling evidence that one should change their dating behavior?

# Let's Talk About P-values

- **A consequence:** People conflate *statistical significance* with *practical significance*.
- We won't use the “**statistically significant**” language in DATA 100. Instead say “**statistically discernible**.”



# Let's Talk About P-values

- Despite its issues, p-values are still quite popular and can still be a useful tool **when used properly**.
- In 2014, George Cobb a professor from Mount Holyoke College posed the following questions (and answers):

Q: Why do so many colleges and grad schools teach  $p = 0.05$ ?

A: Because that's still what the scientific community and journal editors use.

Q: Why do so many people still use  $p = 0.05$ ?

A: Because that's what they were taught in college or grad school.

- I want us to stop this cycle.

# DATA 100 & P-Values

- Understanding p-values and being able to **interpret a p-value in context** is a learning objective of DATA 100.
  - Ex: If ESP doesn't exist, the probability of guessing correctly on at least 106 out of 329 trials is 0.003.
- Understanding that a small p-value means we have evidence for  $H_a$  is important.
  - Ex: Because the p-value is small, we have evidence for ESP.
- Understanding that a small p-value alone does not imply practical significance.
  - Create a confidence interval to measure the effect size!
- Understanding that what you mean by **small** should depend on your field and whether a Type I Error or Type II Error is worse for **your particular research question**.
- Your ability to tell if a # is less than 0.05 is not a learning objective for DATA 100.

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  - Orals throughout the day on Fri (April 10th). **In-person** this time!

